Categorical logic

On course webpage:

- notes
- links to videos
- external material

Assesment

- presentations in the seminar slot
- oral exam

Algebraic theories

Perspectives

- Algebraic structure on sets vs. algebraic structure on „spaces"
  In what generality can we interpret algebraic structure
- When are two notions of algebra equivalent
- Categorification: presenting mathematical structures thru the lens of chategory theory
  E.g. algebraic theories as lawvere theories
  categories of algebras as functor cathegory (functional semantics)

A topological group is a ~~set~~ top. space G with continious $m: G \times G \to G$ (topological product),

$e \in G$ $(G^0 \to G)$, $i: G \to G$

$m(x, m(y,z)) = m(m(x,y), z)$

$m(e, x) = x = m(x, e)$

$m(i(x), x) = m(x, i(x)) = e$

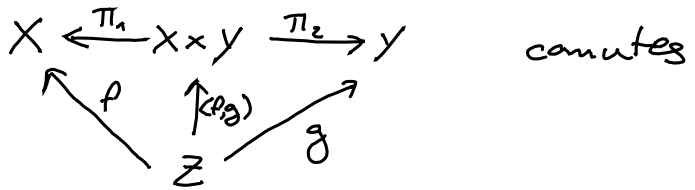
More generaly we can define a group object in a chategory $\mathbb{C}$ in any chategory $\mathbb{C}$ with finite products

A chategery with final products has

- binary products: for any ob X, y we have

$$X \xleftarrow{\pi_1} X \times y \xrightarrow{\pi_2} y$$

such that for any span $(X \xleftarrow{f} z \xrightarrow{g} y)$ there exists $\exists!\ z \xrightarrow{(f,g)} X \times y$ such that



commutes

- terminal object 1
for any object $X \in \mathbb{C}$ there exists a unique map from $X \to 1$

A group object in a category $\mathbb{C}$ with finite products is an object G plus morphisms

$$m : G \times G \longrightarrow G$$

$$e : 1 \longrightarrow G$$

$$i : G \longrightarrow G$$

satisfying

$$\begin{array}{ccccc} G\times(G\times G) & & \xrightarrow{G\times m} & & G\times G \\ \downarrow \cong & & & & \downarrow m \\ (G\times G)\times G & \xrightarrow[m\times G]{} & G\times G & \xrightarrow[m]{} & G \end{array}$$

where $\cong$ is $((\pi_1, \pi_1\circ\pi_2), \pi_2\circ\pi_2)$

$$\begin{array}{ccc} G & \xrightarrow{(e\circ !_G,\, id_G)} & G \\ & \searrow^{id_G} & \downarrow \\ & & G \end{array}$$

+2 for inverse

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow^{z} & \\ & & \end{array}$$

$$Z\times X \xrightarrow[Z\times f]{(\pi_1, f\circ\pi_1)} Z\times Y$$

The category $\mathrm{Grp}_{\mathbb{C}}$ of group objects in $\mathbb{C}$ is given by:

- Objects: Group objects in $\mathbb{C}$ (G, m, e, i)
- Morphisms $(G, M, e, i) \xrightarrow{h} (G', M', e', i')$ in $\mathbb{C}$ such that

$$\begin{array}{ccc} G \times G & \xrightarrow{h \times h} & G' \times G' \\ {\scriptstyle m}\downarrow & & \downarrow{\scriptstyle m} \\ G & \longrightarrow & G' \end{array}$$

$$\begin{array}{ccc} & 1 & \\ {\scriptstyle e}\swarrow & & \searrow{\scriptstyle e'} \\ G & \xrightarrow[h]{} & G' \end{array}$$

$$\begin{array}{ccc} G & \xrightarrow{h} & G' \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle i'} \\ G' & \longrightarrow & G' \end{array}$$

- identities $(G, m, i, e) \xrightarrow{id_G} (G, m, i, e)$
- composition $(G, m, i, e) \xrightarrow{h} (G', m', i', e') \xrightarrow{h'} (G'', m'', i'', e')$, with $h' \circ h$ in $\mathbb{C}$

An algebraic signature Σ is a collective of operations, where each operation f has an associated arity $k \in \mathbb{N}$ ($\mathbb{N}$ includes 0)

The terms T are built by:

- variables x are terms
- if $t_1 \dots t_k$ are terms and f an operation of arity k, then $f(t_1 \dots t_k)$ is a term.

Notation $t(x_1, \dots, x_n)$ to mean all variables in t contained in the ~~left~~ set $\{x_1 \dots x_n\}$

Equation: A pair of terms (t, t') written

$$t = t_1 \quad (x_1, \dots, x_n)$$

Algebraic theory $T = \Sigma + \mathcal{E}$

(Σ: signature; $\mathcal{E}$: collection of equations)

A T-object in $\mathbb{C}$ (with final product) is:

- an object A of $\mathbb{C}$
- for every operation f in $f \in \Sigma$ of arity k we have a morphism $A^k \xrightarrow{f^A} A$

$$A^0 := 1$$
$$A^1 := A$$
$$A^2 := A \times A$$
$$A^{n+1} = A^n \times A$$

such that all equations in $\mathcal{E}$ are true in A

We interpret any term $t\ (x_1 \dots x_n)$ ($x_1 \dots x_n$ distinct) as a morphism $A^n \xrightarrow{t^A} A$

$$A^n \xrightarrow{x_i} A := A^n \xrightarrow{\pi_i} A$$

$$A^n \xrightarrow{f(t_1 \dots t_k)^A} A := A^n \xrightarrow{(t_1^A \dots t_k^A)} A^k \xrightarrow{f^A} A$$

This gives us an interpretation of all terms

An equation $t = t'$ is satisfied if

$$A^n \xrightarrow{t^A} A = A^n \xrightarrow{(t')^A} A$$

The Category $\mathrm{Mod}(T, \mathbb{C})$ of T-objects in $\mathbb{C}$ is defined by

- Objects: T-objects in $\mathbb{C}$ $(A, (f^A)_{f \in \Sigma})$
- Morphisms $(A, (f^A)_{f\in\Sigma}) \to (B, (f^B)_{f\in\Sigma})$ morphisms $A \xrightarrow{h} B$ s.t. $\forall f \in \Sigma$ of arity k

$$\begin{array}{ccc} A^k & \xrightarrow{h^k} & B^k \\ {\scriptstyle f^A}\downarrow & & \downarrow{\scriptstyle f^B} \\ A & \xrightarrow[h]{} & B \end{array}$$

The Lawvere theory $\mathbb{T}$ of an algebraic theory $T = (\Sigma, \mathcal{E})$

$\mathbb{T}$ objects: $\underline{n}$ $n \in \mathbb{N}$

morphisms: $\underline{m} \longrightarrow \underline{n}$ $\qquad \underline{n} = \underline{1}^n$

are n-tuples of terms where $([t_1], \ldots [t_n])$ ← equivalence classes of terms modulo provable equality from equations

$t_1, \ldots t_n \quad (x_1 \ldots x_m)$

identity $\underline{n} \longrightarrow \underline{n} := \underline{n} \xrightarrow{([x_1] \ldots [x_n])} \underline{n}$

composition $\underline{l} \xrightarrow[(x_1 \ldots x_l)]{([t_1], \ldots [t_m])} \underline{m} \xrightarrow[(x_1 \ldots x_m)]{([t_1'], \ldots [t_n'])} \underline{n}$

$$\underline{l} \xrightarrow[(x_1, \ldots x_l)]{([t_1'[t_1 \ldots t_m / x_1 \ldots x_m]], \ldots)} \underline{n}$$

Two algebraic theories T, T' are equivalent if corresponding lawvere theories are equivalent

- category
- equivalence of category
- functors
- natural transformations

$\mathbb{T}$ has final product:

Terminal object $\underline{0}$

$\underline{m} \times \underline{n} = \underline{m+n}$

Theorem: $Mod(\mathbb{T}, \mathbb{C}) \simeq FP[\mathbb{T}, \mathbb{C}]$

(FP: final product preserving functors)

objects:

arrows: natural transformation

A functor $\mathbb{C} \longrightarrow \mathbb{D}$ (categories with final product) preserves final product if
- $F(1_{\mathbb{C}})$ is terminal object
- $F(X \times Y)$

($F(1_{\mathbb{C}}) = 1_{\mathbb{D}}$

$F(X \times Y) = F(X) \times F(Y)$ ... preserving product on the nose)

$FX \xleftarrow{F(\pi_1)} F(X \times Y) \xrightarrow{F(\pi_2)} FY$ is a product diagram in $\mathbb{D}$